

Experiment No : 03

Experiment Name: Determination of the frequency of a tuning fork by Melde's Apparatus

Theory:

In Melde's experiment, one end of a light string is attached to a prong of a tuning fork and the other end passes over a pulley to a scale pan, which provides tension. When the tuning fork is set into vibration, waves travel along the string and are reflected at the fixed end, producing stationary waves with nodes and anti-nodes.

The vibration may be transverse or longitudinal depending on the orientation of the tuning fork. In the transverse arrangement, the plane of vibration is perpendicular to the string and the frequency of the tuning fork N is equal to the frequency of the string n . In the longitudinal arrangement, the plane of vibration is parallel to the string and the frequency of the tuning fork is twice that of the string, $N = 2n$.

Transverse arrangement;

$$N = n = \frac{1}{\lambda} \sqrt{\frac{T}{m}} \dots (i)$$

Longitudinal arrangement;

$$N = 2n = \frac{2}{\lambda} \sqrt{\frac{T}{m}} \dots (ii)$$

Wavelength of the stationary wave, $\lambda = \frac{2l}{p}$

Tension in the string, $T = (w + w_1)g$
.....(iii)

Where,

m = mass per unit length of the string

l = vibrating length

p = number of loops

w = mass of scale pan

w_1 = added mass

g = acceleration due to gravity

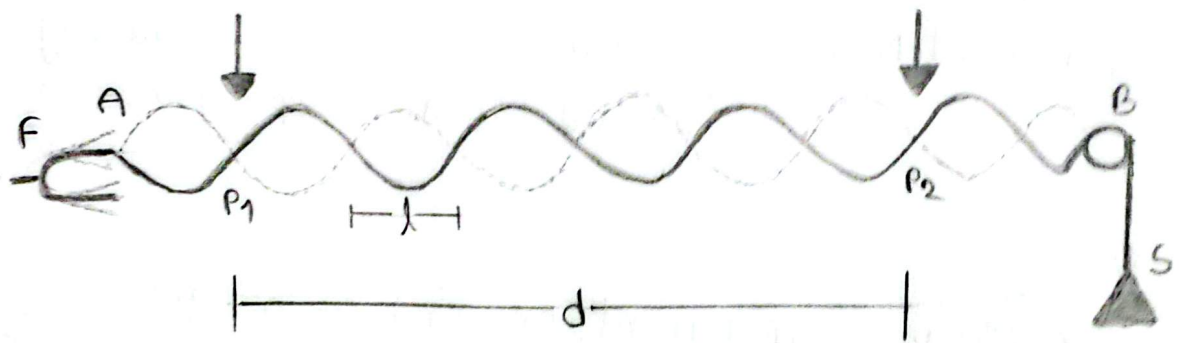


Figure 1: Transverse arrangement

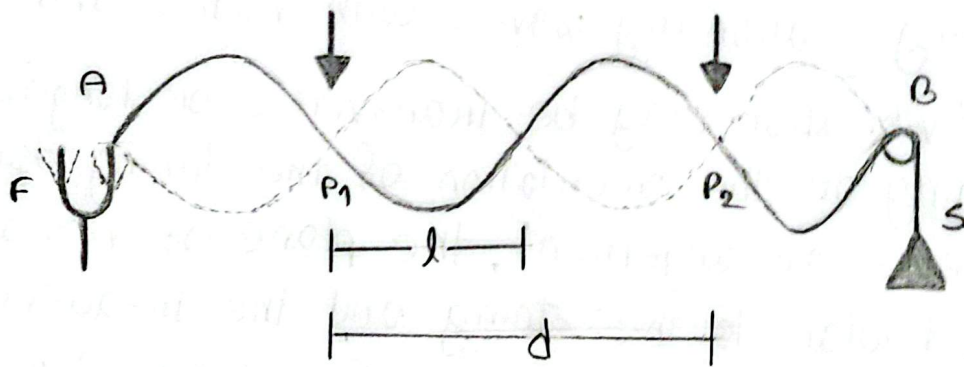


Figure 2: Longitudinal arrangement

Required Apparatus:

- Melde's Apparatus
- String
- Weight box
- Balance

Calculations:

Experimental Data:

Table 01 - Data for frequency of the transverse arrangement

No. of obs.	Mass on the scale pan, w_1 (gm)	Tension, $T = (w + w_1)g$ (dynes)	Total no. of loops (k) between the points P_1 and P_2	Distance between the points P_1 and P_2 , d (cm)	Wavelength $\lambda = \frac{2d}{k}$ (cm)	Mass per unit length of the string m ($\frac{\text{gm}}{\text{cm}}$)	Frequency of the string, $n = \frac{1}{\lambda} \sqrt{\frac{T}{m}}$ (Hz)	Frequency of the fork, $N = n$ (Hz)
1								
2								
3								

Table 02- Data for frequency of the longitudinal arrangement

No. of obs.	Mass on the scale pan, w_1 (gm)	Tension, $T = (w + w_1)g$ (dynes)	Total no. of loops (k) between the points P_1 and P_2	Distance between the points P_1 and P_2 , d (cm)	Wavelength $\lambda = \frac{2d}{k}$ (cm)	Mass per unit length of the string m ($\frac{\text{gm}}{\text{cm}}$)	Frequency of the string $n = \frac{1}{\lambda} \sqrt{\frac{T}{m}}$ (Hz)	Frequency of the fork $N = 2n$ (Hz)
1								
2								
3								

Results:

Precautions:

- Fractional loop was avoided. To get complete loop, the load on the scale pan was increased or decreased or the distance between the points A and B was adjusted
- The length between the pulley and the scale pan was kept as short as possible to get the correct tension
- The total number of loops was increased or decreased by decreasing or increasing the load on the scale pan